

NAME _____



GOSFORD HIGH SCHOOL

2015

Preliminary

Higher School Certificate

MATHEMATICS

Extension 1

Assessment Task 2

General Instructions

- Reading time 5 minutes
- Working time 90 minutes
- Write using black or blue pen
- Board approved calculators may be used
- Use multiple choice answer sheet provided for Questions 1 to 7
- For Questions 8 to 10, show relevant mathematical reasoning and/or calculations
- Total marks 68

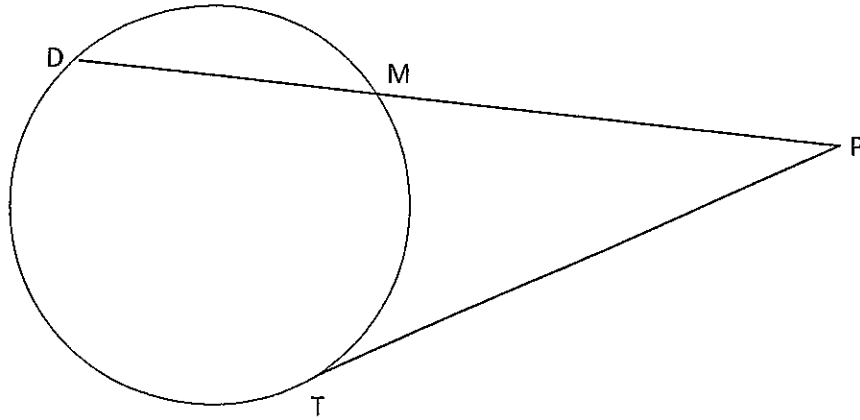
Question 1

$$\tan 2A = ?$$

- (A) $\frac{2\tan A}{1+\tan A}$ (B) $\frac{2\tan A}{1-\tan A}$ (C) $\frac{2\tan A}{1+\tan^2 A}$ (D) $\frac{2\tan A}{1-\tan^2 A}$

Question 2

PT is a tangent to the circle and PD is a secant.

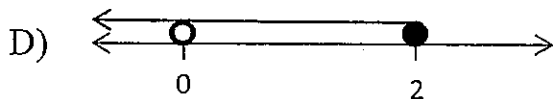
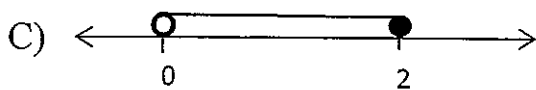
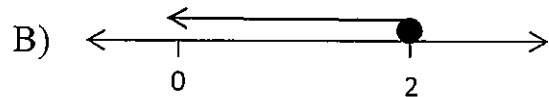
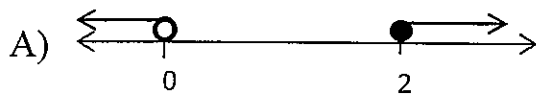


Therefore $PT^2 = ?$

- (A) $PM \times MD$ (B) $PM \times PD$ (C) $DM \times DP$ (D) $PM + MD$

Question 3

Which of the following represents the graphical solution to the inequality $\frac{2}{x} \geq 1$



Question 4

A zookeeper has to accommodate 7 different animals in 7 enclosures, one in each enclosure. However, 2 of the animals are too large for 3 of the enclosures.

The number of different possible arrangements of the animals is :

- (A) 5040 (B) 240 (C) 1440 (D) 720

Question 5

Three students write down different expressions for $\sin^2 x$

Lena's answer states that $\sin^2 x = 1 - \cos^2 x$

Will's answer states that $\sin^2 x = 4\sin^2\left(\frac{x}{2}\right) \cdot \cos^2\left(\frac{x}{2}\right)$

Abby's answer states that $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$

Which of the following statements is correct

- | | |
|-----------------------------------|-----------------------------------|
| A) Only Lena is correct | B) Only Lena and Will are correct |
| C) Only Lena and Abby are correct | D) All three students are correct |

Question 6

For all values of X and Y

$$\cos X \cdot \cos(X + Y) + \sin X \cdot \sin(X + Y)$$

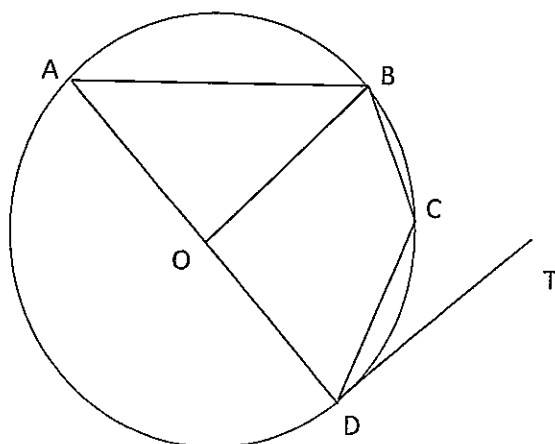
is equal to

- (A) $\cos X$ (B) $\cos Y$ (C) $-\cos Y$ (D) $\cos(2X + Y)$

Question 7

O is the centre of the circle below. A, B, C and D all lie on the circle. AD is a diameter, TD is a tangent, $\angle TDC = 38^\circ$ and $\angle DCB = 108^\circ$.

DIAGRAM
NOT
DRAWN TO
SCALE



$\angle CBO = ?$

- A) 56° B) 52° (C) 128° (D) 72°

Question 8 (19 marks)

- (a) Simplify $\frac{9^n - 1}{3^n - 1}$ (2)
- (b) Factorise $y^2 - x^2 + 2x - 1$ (2)
- (c) Simplify $\frac{yx^{-1} - xy^{-1}}{x+y}$ (2)
- (d) Solve $\frac{1}{x+2} < 1$ (3)
- (e) A committee of three is to be formed from 4 men and 5 women.
Find the number of different selections are possible if :
- (i) there are no other restrictions on selection (1)
- (ii) there is to be at least one man on the committee? (2)
- (f) In how many ways can the letters of the word READER be arranged if :
- (i) there are no restrictions on the arrangements (1)
- (ii) the two R's are together (1)
- (g) In how many ways can 5 men and 5 women be arranged in a circle if
- (i) there is no restrictions (1)
- (ii) no two men are to sit next to each other (1)
- (h) Solve $\frac{1}{(x-1)(x-3)} < -1$ (3)

Question 9

(22 marks)

- (a) Find the exact value of $\sin 105^\circ$ (2)
- (b) Express $\sin 2x \cdot \cot x - 1$ in simplest form (3)
- (c) Simplify $\tan(x + 45^\circ) \cdot \tan(x - 45^\circ)$ (2)
- (d) Solve $\sin \alpha = \cos 2\alpha$ for $0^\circ \leq \alpha \leq 360^\circ$ (3)
- (e) If $t = \tan\left(\frac{\emptyset}{2}\right)$,
- (i) express $\sin \emptyset$ and $\cos \emptyset$ in terms of t (2)
- (ii) hence, solve $\cos \emptyset + \sin \emptyset + 1 = 0$ for $0^\circ \leq \emptyset \leq 360$ (3)
- (f) (i) Prove that $\cos 3x = 4\cos^3 x - 3\cos x$ (3)
- (ii) Explain why, if $x = 18^\circ$, then $\sin 2x = \cos 3x$ (1)
- (iii) With consideration to parts (i) and (ii) above ,
- prove that $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$ (3)

Question 10

(20 marks)

- (a) In the diagram, O is the centre of the circle and $\angle BOC = 136^\circ$ as shown. Find $\angle BAC$, giving reasons.

(2)

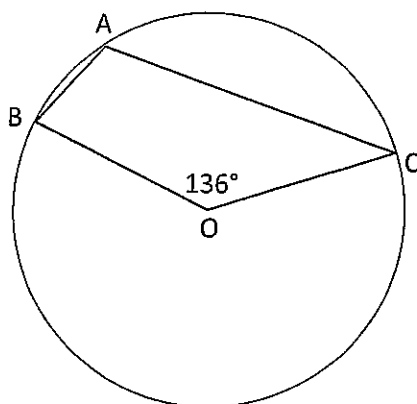


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- (b) Given that RT is a diameter, find x giving reasons.

(3)

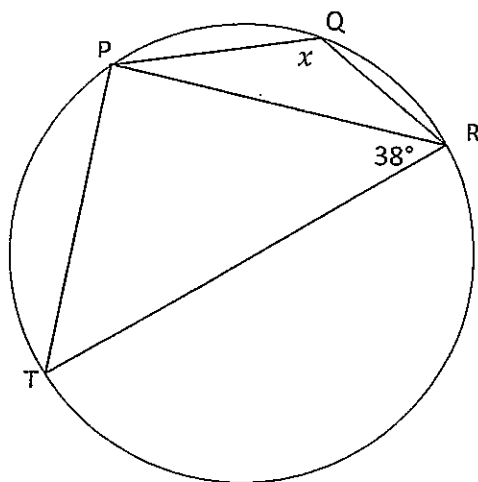


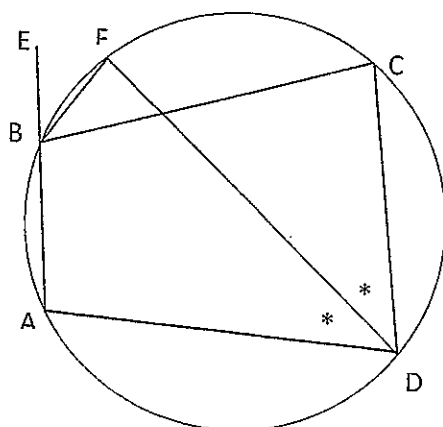
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- (c) A, B, C, D and F all lie on the given circle and AB has been produced to E.

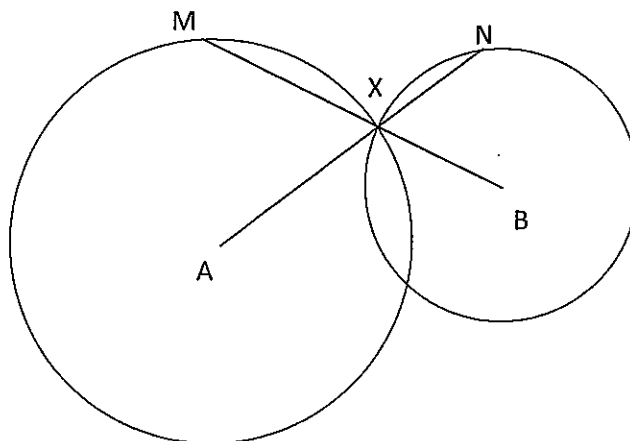
FD bisects angle ADC.

Prove that FB bisects the angle EBC.

(3)

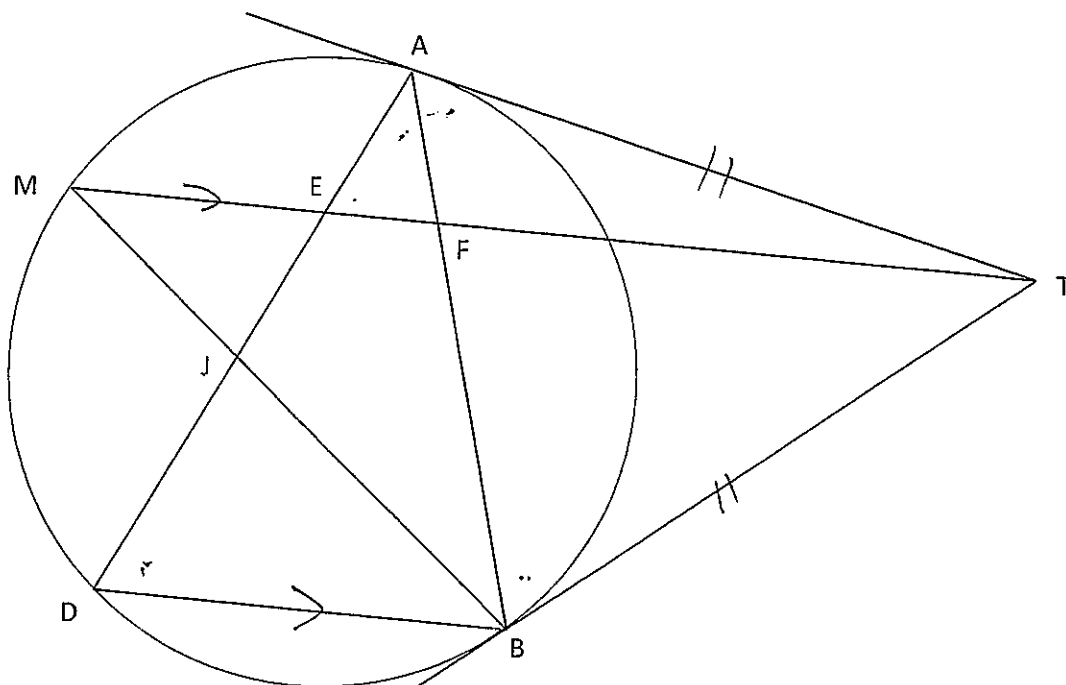


- (d) The two circles have centres A and B.
M and N are points on the circumferences of these respective circles.
AN and BM intersect at X, where X is one of the points of intersection of the two circles.



- (i) Prove $\angle AMX = \angle AXM$ (1)
- (ii) Prove $\angle AMX = \angle BNX$ (2)
- (iii) State why A, B, N and M are concyclic. (1)

- (e) In the diagram below TA and TB are tangents to the circle.
 AB, BD, MB and DA are all chords.
 MT is a secant which is parallel to the chord BD.



- (i) Give reasons why triangle TAB is isosceles. (2)
- (ii) State why $\angle TAB = \angle AET$ (2)
- (iii) Prove $\triangle TAE$ is similar to $\triangle TFA$ (2)
- (iv) Prove $TB^2 = TE \cdot TF$ (2)

End of Examination

NAME _____

ANSWER SHEET FOR SECTION 1 - Multiple Choice

1.	A	B	C	D
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2. A B C D

3. A B C D

4. A B C D

5. A B C D

6. A B C D

7. A B C D

(1) D (2) B (3) C (4) C (5) D (6) B (7) A

QUESTION 8

$$\begin{aligned}
 (a) \quad \frac{9^n - 1}{3^n - 1} &= \frac{(3^2)^n - 1}{3^n - 1} \\
 &= \frac{(3^n)^2 - 1}{3^n - 1} \\
 &= \frac{(3^n - 1)(3^n + 1)}{3^n - 1} && 1 \text{ mark} \\
 &= 3^n + 1 && 1 \text{ mark}
 \end{aligned}$$

$$\begin{aligned}
 b) \quad y^2 - x^2 + 2x - 1 &= y^2 - (x^2 - 2x + 1) \\
 &= y^2 - (x - 1)^2 && 1 \text{ mark} \\
 &= (y - x + 1)(y + x - 1) && 1 \text{ mark}
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad \frac{yx^{-1} - xy^{-1}}{x+y} &= \frac{\frac{y}{x} - \frac{x}{y}}{x+y} \\
 &= \frac{y^2 - x^2}{xy(x+y)} && 1 \text{ mark} \\
 &= \frac{(y-x)(y+x)}{xy(x+y)} && 1 \text{ mark} \\
 &= \frac{y-x}{xy} && 1 \text{ mark}
 \end{aligned}$$

$$(e) \quad (i) \quad \text{N}^\circ \text{ of selections} = {}^9C_3 \quad (1)$$

$$\begin{aligned}
 (ii) \quad \text{N}^\circ \text{ of selections} &= {}^4C_1 \times {}^5C_2 + {}^4C_2 \times {}^5C_1 + {}^4C_3 \\
 &= \quad \quad \quad (2)
 \end{aligned}$$

$$(f) \quad (i) \quad \text{N}^\circ \text{ of arrangements} = \frac{6!}{2!2!} = 180 \quad (1)$$

$$(ii) \quad \text{N}^\circ \text{ of arrangements} = \frac{1 \times 5!}{2!} = 60 \quad (1)$$

$$(g) \quad (i) \quad \text{N}^\circ \text{ of arrangements} = 9! \quad (1)$$

$$(ii) \quad \text{N}^\circ \text{ of arrangements} = 4!5! \quad (1)$$

(h) Critical values when $x=1, 3$ and when

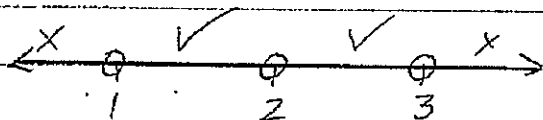
$$\frac{1}{(x-1)(x-3)} = -1 \quad \Rightarrow \quad -1 = (x-1)(x-3)$$

$$-1 = x^2 - 4x + 3$$

$$x^2 - 4x + 4 = 0$$

$$(x-2)^2 = 0$$

$$x = 2$$



Test $x < 1$; (say $x=0$); $\frac{1}{3} < -1$, False

Test $1 < x < 2$; (say $x=\frac{3}{2}$); $-\frac{4}{3} < -1$; True.

Test $2 < x < 3$ (say $x=\frac{5}{2}$); $-\frac{4}{3} < -1$; True

Test $x > 3$ (say $x=4$); $\frac{1}{3} < -1$; False

$$\therefore 1 < x < 2 \text{ or } 2 < x < 3$$

Question 9

$$\begin{aligned} (a) \quad \sin(105)^\circ &= \sin(45 + 60)^\circ \\ &= \sin 45 \cos 60 + \sin 60 \cos 45^\circ \quad (1) \\ &= \frac{1}{\sqrt{2}} \times \frac{1}{2} + \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} \quad (1) \\ &= \frac{1 + \sqrt{3}}{2\sqrt{2}} \end{aligned}$$

$$\begin{aligned} (b) \quad \sin 2x \cdot \cot x - 1 &= 2 \sin x \cos x \cdot \frac{\cos x}{\sin x} - 1 \quad (1) \\ &= 2 \cos^2 x - 1 \quad (1) \\ &= \cos 2x \quad (1) \end{aligned}$$

$$\begin{aligned} (c) \quad \tan(x + 45^\circ) \cdot \tan(x - 45^\circ) &= \left(\frac{\tan x + \tan 45^\circ}{1 - \tan x \cdot \tan 45^\circ} \right) \cdot \frac{\tan x - \tan 45^\circ}{1 + \tan x \tan 45^\circ} \\ &= \frac{\tan x + 1}{1 - \tan x} \cdot \frac{\tan x - 1}{1 + \tan x} \quad (1) \\ &= -1 \quad (1) \end{aligned}$$

$$\begin{aligned} (d) \quad \sin \alpha &= 1 - 2\sin^2 \alpha \quad (1) \\ 2\sin^2 \alpha + \sin \alpha - 1 &= 0 \\ (2\sin \alpha - 1)(\sin \alpha + 1) &= 0 \quad (1) \\ \sin \alpha &= \frac{1}{2}, -1 \\ \therefore \alpha &= 30^\circ, 150^\circ, 270^\circ \quad (1) \end{aligned}$$

$$(e) \text{ (i) } \sin \phi = \frac{2t}{1+t^2}, \quad \cos \phi = \frac{1-t^2}{1+t^2} \quad (2)$$

$$(ii) \quad \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2} + 1 = 0 \quad \text{Note } 0 \leq \tan \frac{\phi}{2} \leq 180^\circ$$

$$\therefore 1-t^2 + 2t + 1+t^2 = 0 \quad (1)$$

$$2t + 2 = 0$$

$$t = -1 \quad (1)$$

$$\therefore \frac{\phi}{2} = 135^\circ$$

$$\therefore \phi = 270^\circ \text{ and } 180^\circ \text{ (inspector)}$$

$$(f) (i) \quad \cos 3x = \cos(2x+x)$$

$$= \cos 2x \cos x - \sin 2x \sin x \quad (1)$$

$$= (2\cos^2 x - 1) \cos x - 2\sin x \cos x \cdot \sin x$$

$$= 2\cos^3 x - \cos x - 2\cos x \cdot \sin^2 x$$

$$= 2\cos^3 x - \cos x - 2\cos x(1 - \cos^2 x) \quad (1)$$

$$= 2\cos^3 x - \cos x - 2\cos x + 2\cos^3 x$$

$$= 4\cos^3 x - 3\cos x \quad (1)$$

$$(f) (ii) \quad 2x = 36^\circ, \quad 3x = 54^\circ$$

$$\sin 36^\circ = \cos 54^\circ \text{ since } \sin \theta = \cos(90^\circ - \theta) \quad (1)$$

$$(iii) \quad \sin(2 \times 18^\circ) = \cos(3 \times 18^\circ)$$

$$2\sin 18^\circ \cos 18^\circ = 4\cos^3 18^\circ - 3\cos 18^\circ$$

$$2\sin 18^\circ = 4\cos^2 18^\circ - 3$$

$$3 + 2\sin 18^\circ = 4(1 - \sin^2 18^\circ)$$

$$= 4 - 4\sin^2 18^\circ$$

$$4\sin^2 18^\circ + 2\sin 18^\circ - 1 = 0 \quad (1)$$

$$\sin 18^\circ = \frac{-2 \pm \sqrt{4+16}}{8}$$

$$\sin 18^\circ = \frac{-2 \pm \sqrt{20}}{8}$$

$$\sin 18^\circ = \frac{-1 \pm \sqrt{5}}{4} \quad (1)$$

$$\text{But } \sin 18^\circ > 0 \therefore \sin 18^\circ = \frac{-1 + \sqrt{5}}{4}$$

$$(1) \quad 4$$

Question 10

a) $\text{Reflex } \hat{BOC} = 360^\circ - 136^\circ$ (angles at a point form a revolution) (1)
 $= 224^\circ$

$\therefore \hat{BAC} = \frac{1}{2} \times 224^\circ$ (angle at the centre is twice the angle at the circumference standing on the same arc) (1)
 $= 112^\circ$

b) $\hat{TPR} = 90^\circ$ (angle in a semi-circle is a right angle)

$\hat{PTR} = 180^\circ - (90 + 38)^\circ$ (angle sum of a triangle)
 $= 52^\circ$

$x = 180 - 52^\circ$ (opposite angles of a cyclic quadrilateral are supplementary)
 $= 128^\circ$

c) $\hat{EBF} = \hat{ADF}$ (exterior angle of a cyclic quadrilateral is equal to the opposite interior angle)

$\hat{FBC} = \hat{FDC}$ (angles at the circumference standing on the same arc are equal)

But $\hat{ADF} = \hat{FDC}$ (given FD bisects $\hat{ADC}$)

$\therefore \hat{EBF} = \hat{FBC}$ (equal to equal angles)

e) (i) $TA = TB$ (tangents from an external point are equal)

$\therefore \triangle TAB$ is isosceles since triangle has two equal sides. (1)

(ii) $\angle TAB = \angle ADB$ (angle between a tangent and a chord is equal to the angle in the alternate segment)

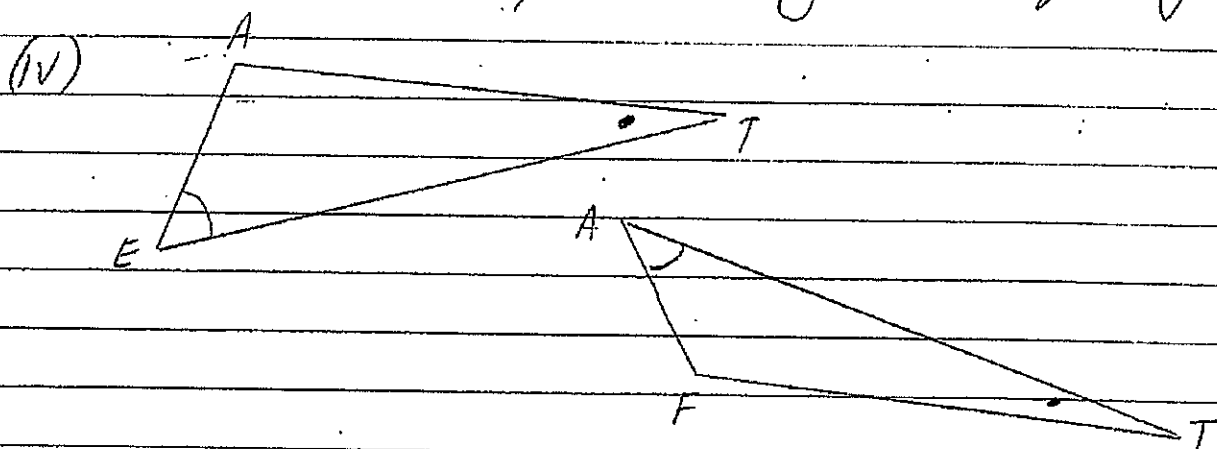
$\angle ADB = \angle AET$ (corresponding angles, $TM \parallel BD$)

$\therefore \angle TAB = \angle AET$ (equal to equal angles) (2)

(iii) $\angle ATE = \angle ATF$ (common angle)

$\angle TAE = \angle AET$ (proven in (ii)) (2)

$\therefore \triangle TAE \sim \triangle TFA$ (triangles are equiangular)



$\frac{TA}{TF} = \frac{TE}{TA}$ (corresponding angles of similar triangles are in proportion)

$\therefore TA^2 = TE \cdot TF$ (2)

$\therefore TB^2 = TE \cdot TF$ (since $TA = TB$)